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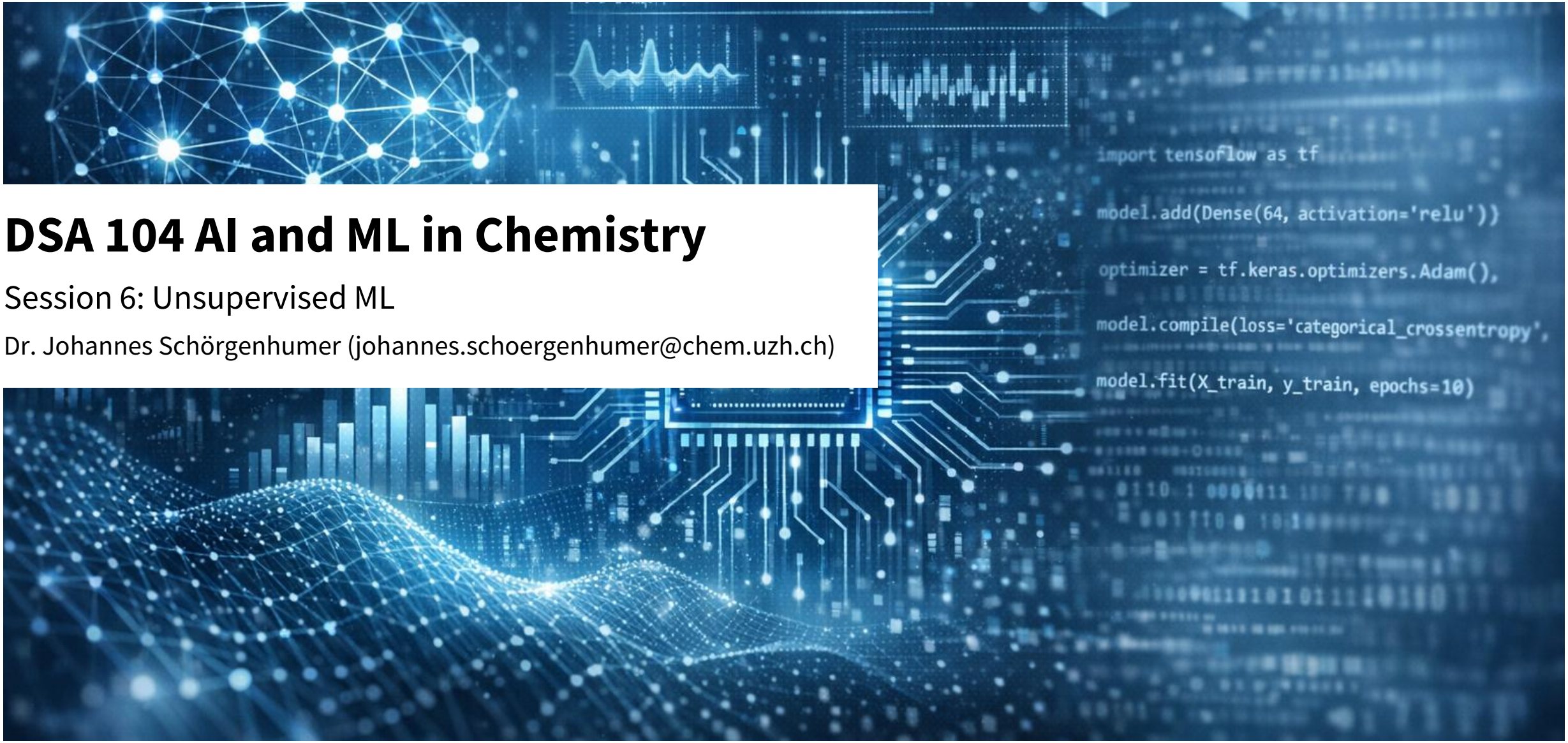

DSA 104 AI and ML in Chemistry

Session 6: Unsupervised ML

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Lesson outline

Wrap up SVC Exercise

Solution: Assignment 1

Types of Unsupervised ML

— Dimensionality reduction

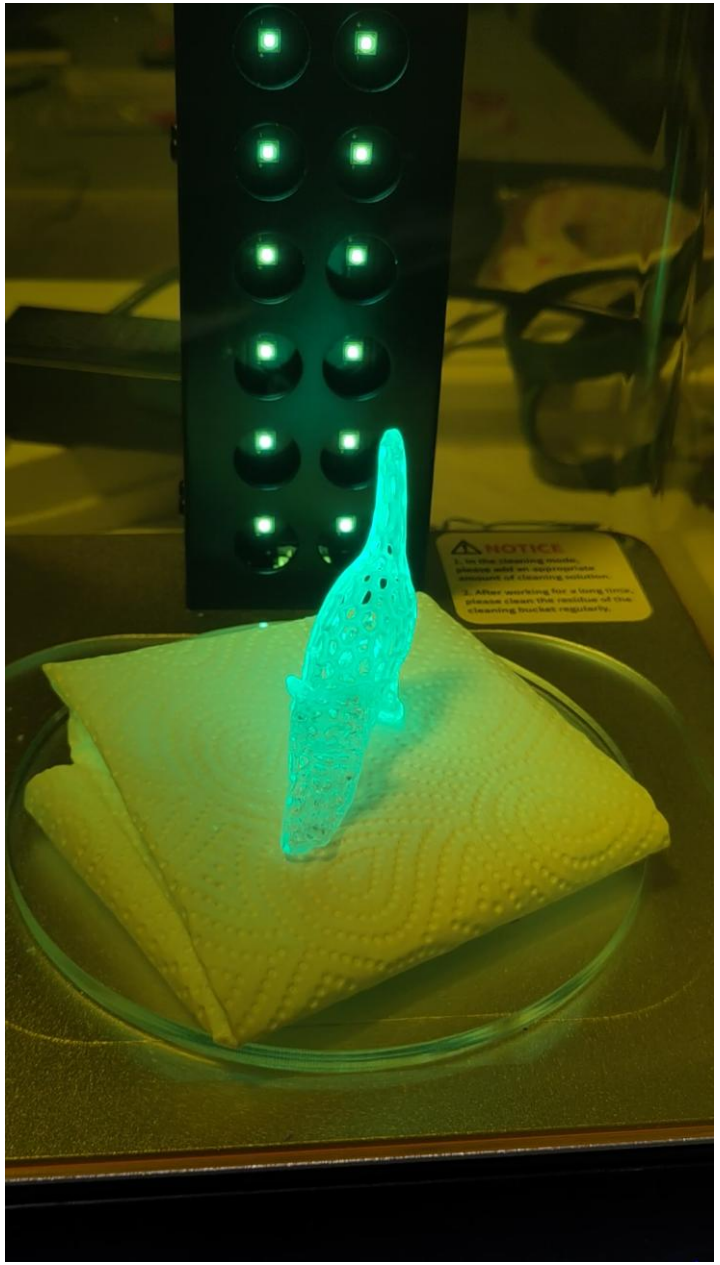
— Clustering

— Density estimation

— Anomaly detection

Data Characteristics and Evaluation of unsupervised ML models

Practical examples



The kernel trick

1) Consider the following minimal dataset:
$x_1 = -2, y_1 = -1$ (class A)
$x_2 = -1, y_2 = -1$ (class B)
$x_3 = 1, y_3 = 1$ (class B)
$x_4 = 2, y_4 = 1$ (class A)

— Plot the data and see if this can be separated in 1D

— Map data using $\phi(x) = (x, x^2)$

— Plot transformed data

— What kernel corresponds to this mapping (assume two points x and z)?

2) Imagine the following kernel $K(x, z) = (1 + xz)^2$. What would be the corresponding feature map (how many dimensions)?

SVC for Breast Cancer set

1) Run a Classification using a SVC with linear kernel on the Wisconsin Breast Cancer dataset (*SVC_breastcancer.ipynb*)
2) Compare different kernels (linear, RBF) and other models on a reduced dataset (more difficult scenario)

In both tasks,

— Use Pipelines

— Perform a GridSearchCV

— Calculate a confusion matrix

Unsupervised ML

- Only input features, no label!!

- Formal framing:
  - Given dataset X
  - Find structure S

- Types of structure:
  - Groups
  - Latent variables
  - Density
  - Low-dimensional representations

Real-World Examples

- Customer segmentation (e.g. e-commerce behavior, Galaxus, Netflix, …)
- Topic discovery in text
- Anomaly detection in fraud
- Data / Image compression

Types of Unsupervised ML: Clustering

Detect patterns in unlabelled data

Clustering = grouping data points such that

— **intra-cluster similarity is high**

— **inter-cluster similarity is low**

Goals:

— Identify **patterns** and **relationships** in complex data

— **Simplify** high-dimensional data into manageable clusters

— **Visualise** and understand underlying structure of the data

— Inform hypothesis generation and experimental design

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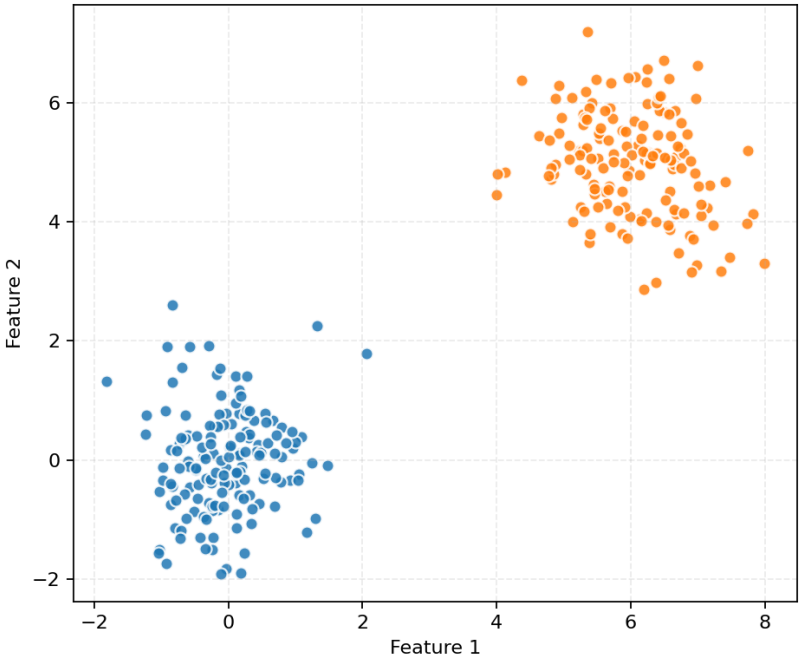
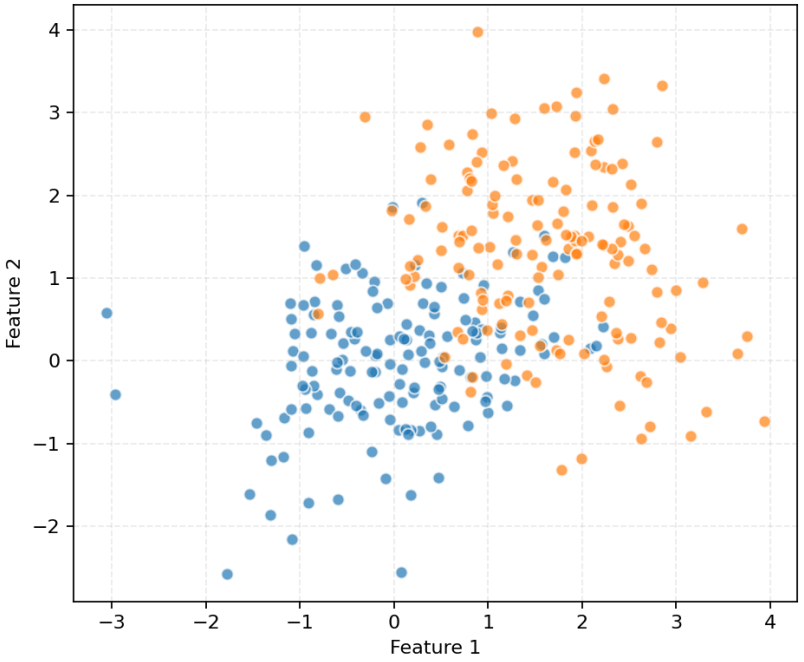
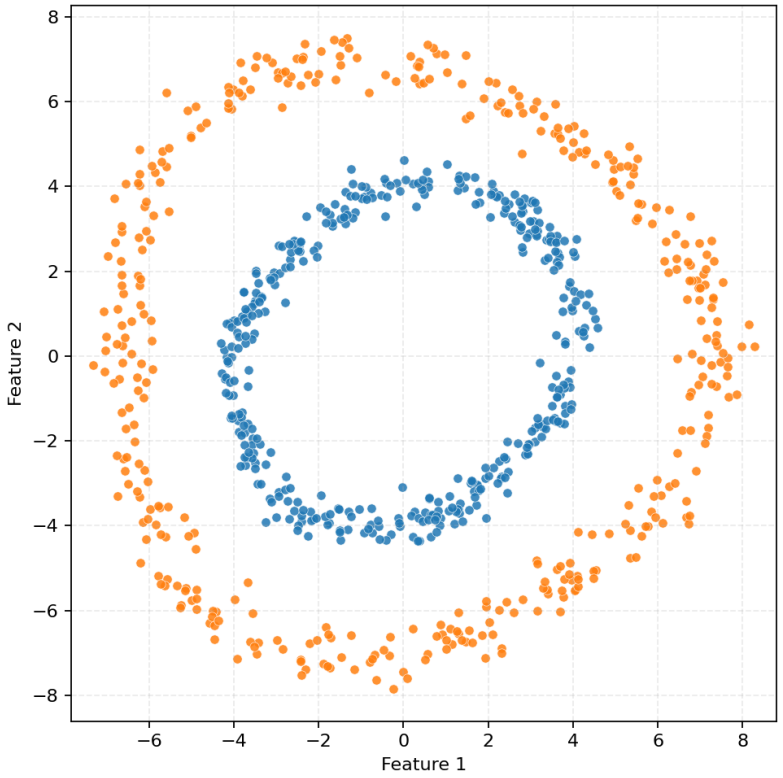
Clustering: What is a cluster?







Clustering depends on:

— **Shape** assumptions (e.g. k-means)

— **Density** assumptions (e.g. DBSCAN)

— **Distance** metric

— **Overlap** allowed (Hard vs. soft clustering!)

$S = f$(Representation, Metric, Assumption)

Different assumptions, different partitions!!

Clustering algorithms – partition-based: K-Means

— Goal: **minimise variance in clusters**

$$\min_{C_1 \dots C_k} \sum_{k=1}^{K} \sum_{x \in C_k} \|x - \mu_k\|^2$$

— Important parameters:

  — K: number of clusters

— Algorithm steps

  — Assign each point to closest centroid

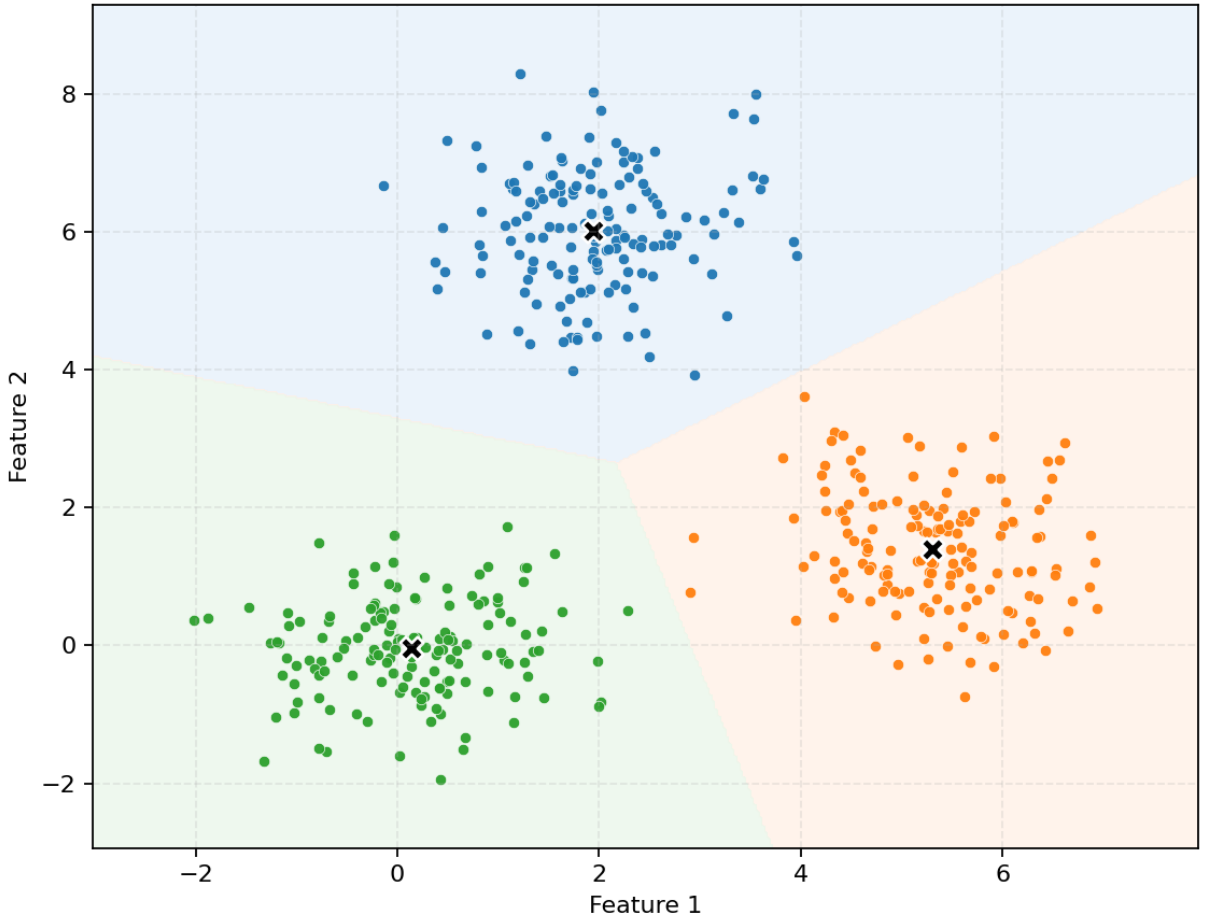
  — Update centroids as cluster means

  — Repeat until converged

— Implicit assumptions:

  — Uses Euclidean distance

  — Uses the mean as representative

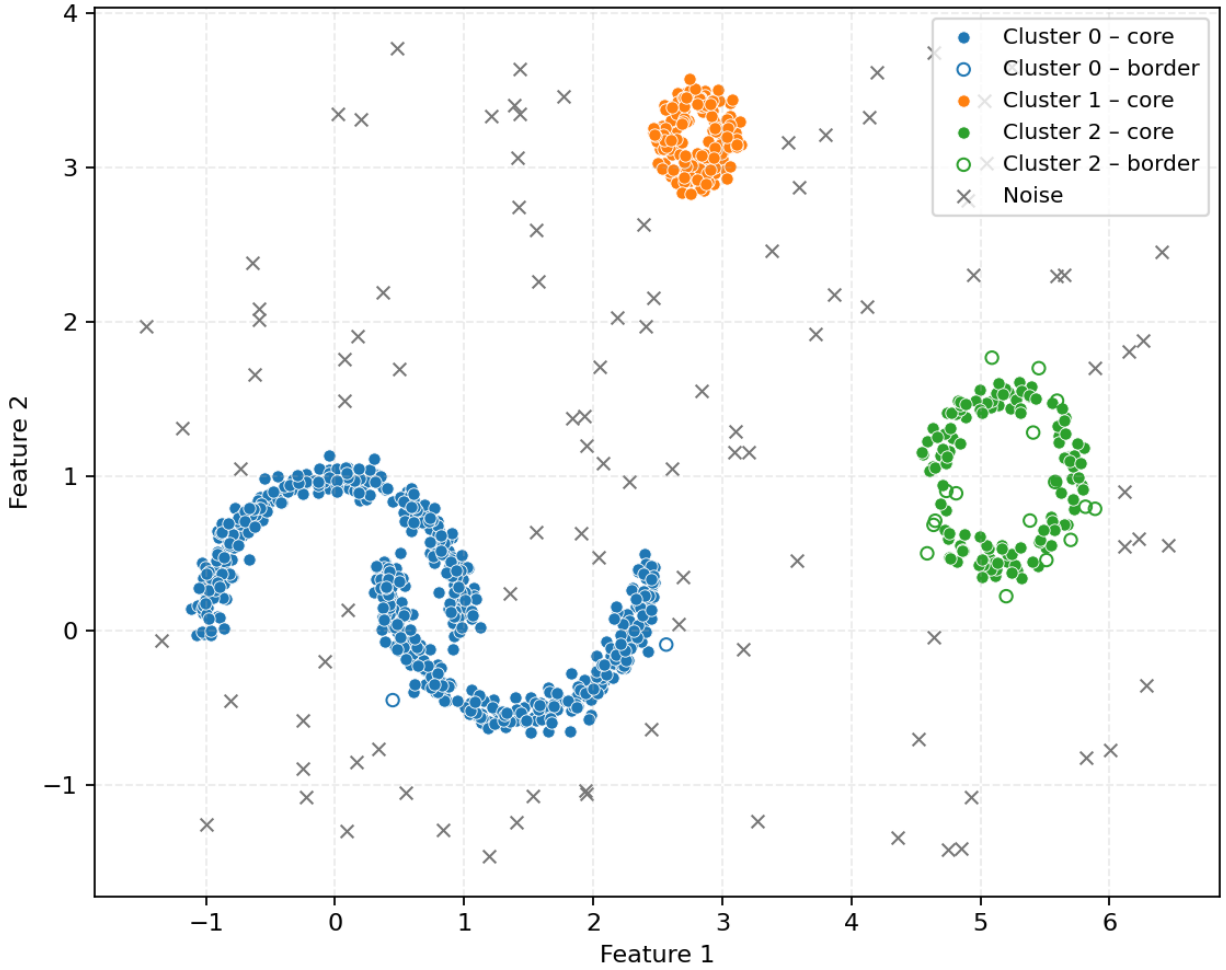


Works well if the clusters are:
— **Spherical** (convex)
— **Similar** size
— Similar density
— Separable in Euclidean space

Usually fails for:
— Non-spherical clusters
— Different densities
— Heavy overlap
— High-dimensional sparse data

Clustering algorithms – density-based: DBSCAN

- Core idea:
  - **Clusters are dense regions**
  - Separated by low-density areas
  - Clusters grow with density connectivity
- Parameters:
  - ε : neighbourhood radius
  - MinPts: minimal number of points to form dense region
- Types of points:
  - Core point ≥ MinPts within ε-neighbourhood
  - Border point: Fewer neighbors, but reachable from a core
  - Noise point: Not density-reachable

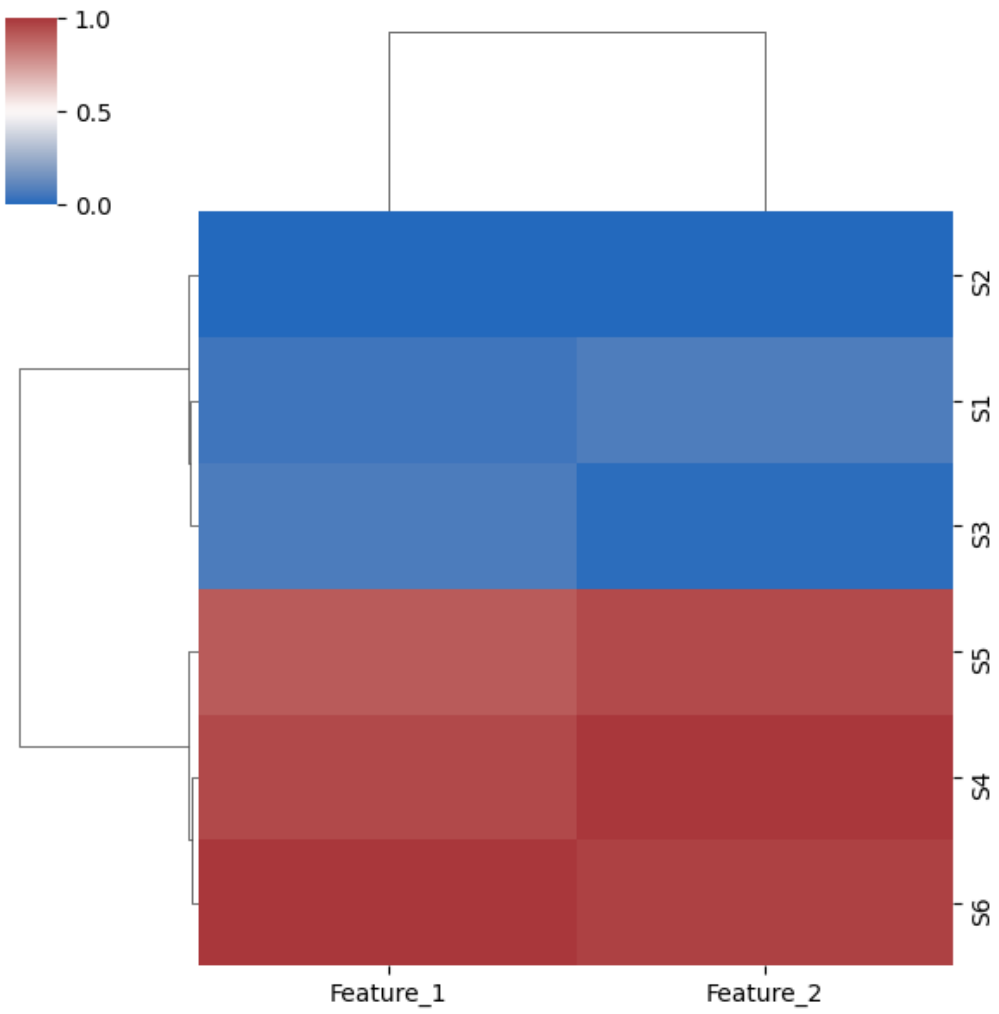


Strengths
- Finds **arbitrary shapes**
- Automatically **detects outliers**
- No need to choose K

Limitations
- Sensitive to ε choice
- Struggles with varying densities
- Hard in high dimensions (distance concentration)

Clustering algorithms: Hierarchical Clustering

- Core idea:
  - Build a **tree of clusters** (dendrogram) by successive **merging** (agglomerative) or splitting (divisive) strategies
  - Height = merge distance, cutting at height h produces K clusters
- Agglomerative (bottom up): iteratively merge closest clusters
- Distance considerations:
  - Euclidean
  - Manhattan
  - Cosine
  - Tanimoto (chemical fingerprints)
- Linkage criteria:
  - Single: minimum pairwise distance
  - Complete: maximum pairwise distance
  - Average: mean distance, overall cohesion
  - Ward linkage: minimises variance increase



Strengths
- **Multi-scale structure**
- No need to pre-specify K
- Works with **arbitrary distance metrics**

Limitations
- Computationally expensive ($O(n^2)$)
- Sensitive to noise
- Early merge decisions can't be undone

Clustering algorithms – Model-based, e.g.: Gaussian Mixture Models

— Core idea / Assumptions:

— Data is generated from **mixture of probability distributions**

$$p(x) = \sum_{k=1}^{K} \pi_k \cdot p(x|\theta_k)$$

Mixing proportion

Cluster distribution with parameters θ_k

— Parametric form, e.g. Gaussian (not only spherical…)

$$p(x \mid k) = \mathcal{N}(x \mid \mu_k, \Sigma_k)$$

Mean vector

Covariance matrix

— **Soft clustering**:

— Uncertainty assessment instead of hard 0/1 assignment

— Expectation-Maximization (EM) algorithm:

— Learning E-step: compute soft assignments $\gamma_{ik} = p(x_i \mid k)$

— M step: Update Σ_k, μ_k, π_k

— Iterate until convergence

— K-Means = a special case of GMM!

Strengths

—Statistically grounded
—Handles **overlapping clusters**
—Can **estimate K** by calculation
—Provides **uncertainty**

Limitations

—Assumes parametric form (e.g. Gaussian)
—Sensitive to initialisation
—Can overfit
—Computationally heavier

Clustering: Exploring Chemical Space

Use the tmap online tool either on the example data provided or on the modified ESOL dataset (*esol_modified.csv*: including molecular descriptors)

https://try-tmap.gdb.tools/

Find more information here: https://tmap.gdb.tools/

Types of Unsupervised ML: Dimensionality reduction

- Linear Methods:
  - PCA: Principal Component Analysis

- Nonlinear Methods:
  - t-SNE: t-distributed Stochastic Neighbour Embedding
  - UMAP: Uniform Manifold Approximation and Projection

Dimensionality reduction: PCA

What PCA is:

— Variance **compression to n** components (maintaining **maximum variance**)

— “Finding the main axes of variation”

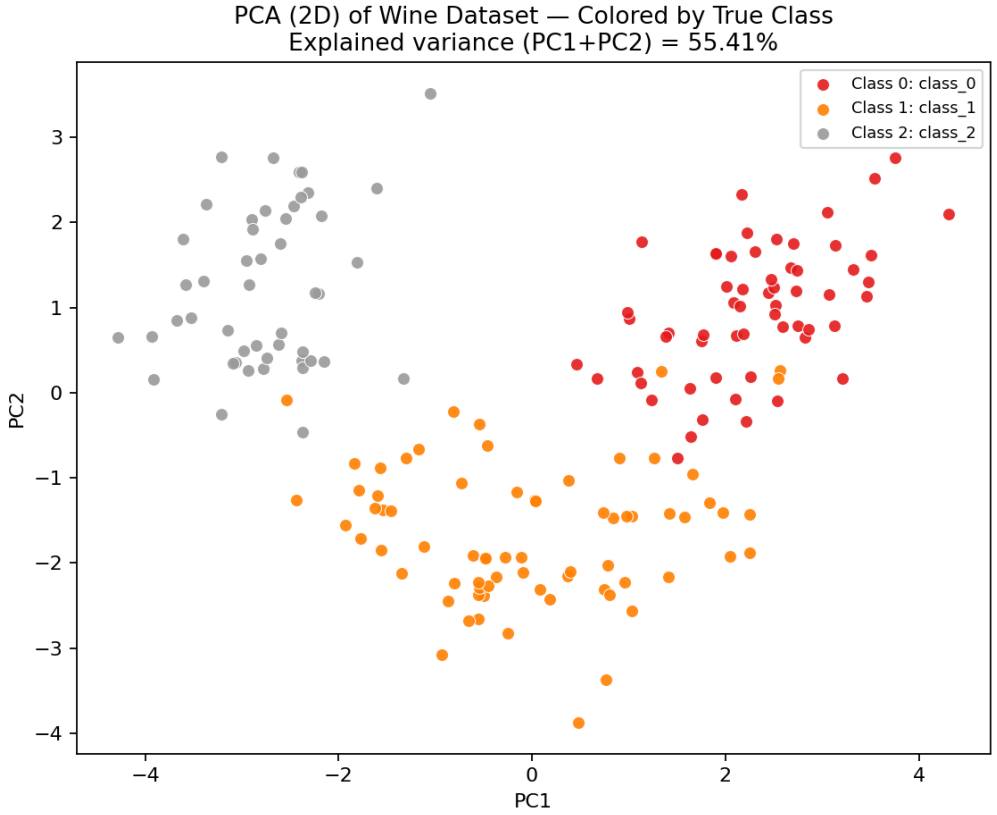
— Linear: **dimensional information retained** (still limited **interpretability**)

What it’s *not*:

— Predictive

— A classifier or clustering technique

— Magic feature selection



Good for:

— High-dimensional datasets

— Data visualisation

— Noise reduction

— Preprocessing for ML models

Working principle:

1) Standardising the data (so each feature has equal importance)

2) Calculate covariance matrix to check how features relate to each other. Finding directions (“principal components”, **PCs**) where the data varies the most.

3) After finding the components that capture the **maximum variance**, the original data is projected onto these new axes

Dimensionality reduction: t-SNE

What it does:

— Preserves **local structure** (nearby points stay nearby)

— Reveals clusters and patterns

— Handles complex, nonlinear relationships

Working principle:

— Convert pairwise distances in high-dimensional space into probabilities

— Do the same in low-dimensional space

— Minimize difference between the two distributions

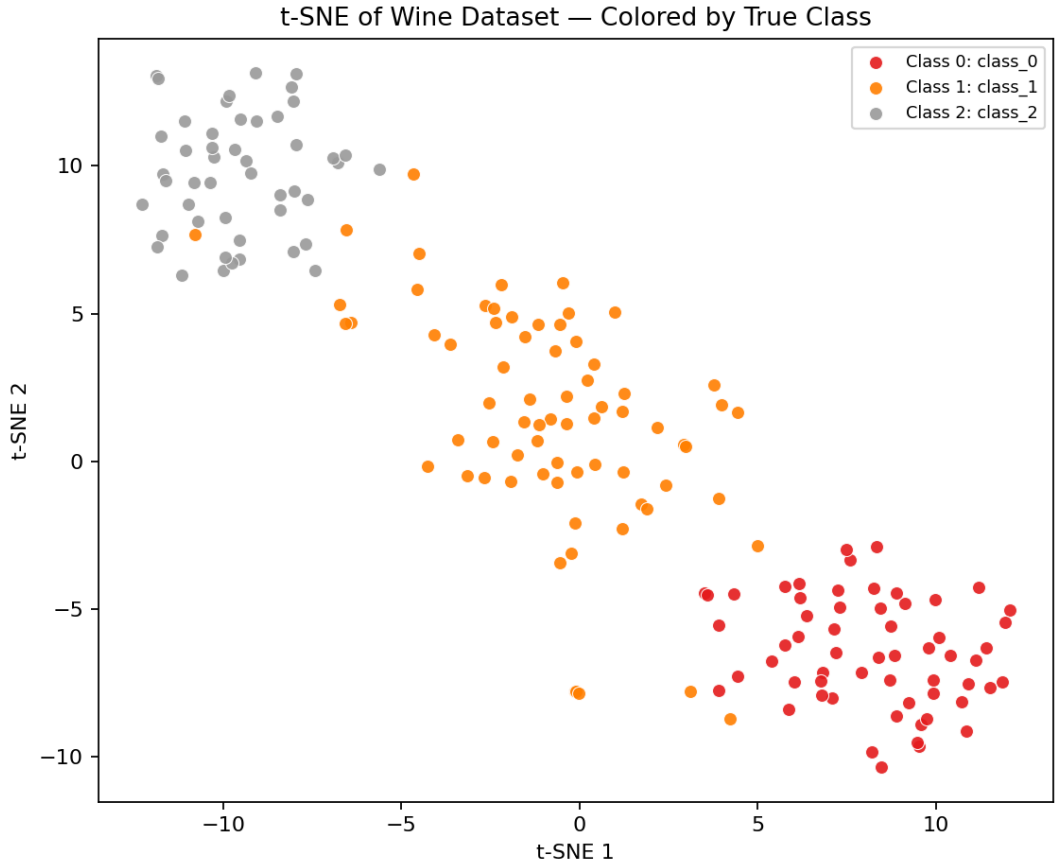
— Uses a Student-t distribution in low-dimensional space to reduce crowding

Key characteristics

— Focuses on local neighborhoods ("**similarity**")

— **Global distances** are **less meaningful**

— Results vary with initialization & hyperparameters (Perplexity, learning rate, number of iterations



Good for:

— Data exploration

— Cluster visualisation

— High-dimensional embeddings (e.g. images, genomics, …)

Dimensionality reduction: UMAP

What it does:

— Preserves **local structure** (like t-SNE), but preserves **global structure** better

— **Scales well** to large datasets

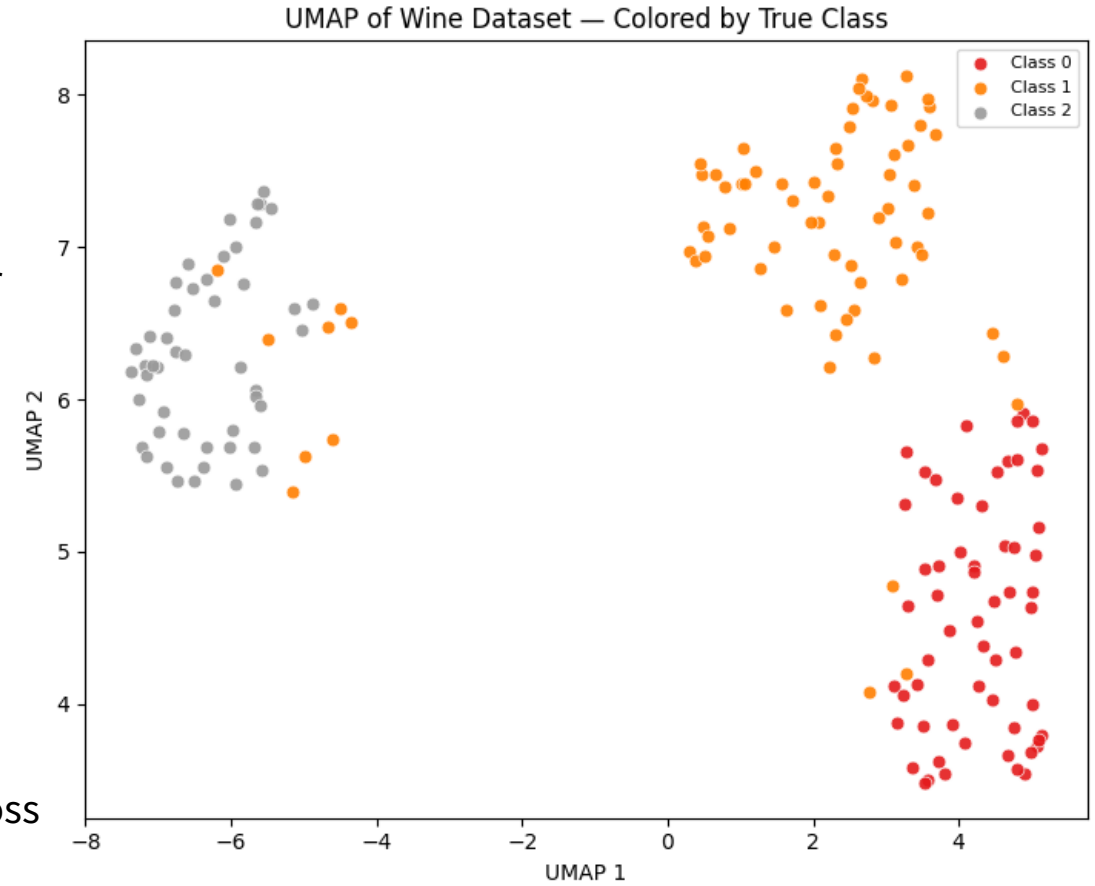
— Faster than t-SNE

Working principle:

— Construct a weighted k-nearest neighbor graph in high-D space

— Model data as a fuzzy topological structure

— Optimise a low-D embedding to preserve this structure→ via cross-entropy loss



Key characteristics

— Balances local + global structure

— Deterministic option (with fixed random seed)

— Produces meaningful embeddings for downstream ML

Good for:

— Large datasets

— **Cluster visualisation**

— Feature extraction for ML

— **Faster, scalable t-SNE-like results**

Important Hyperparameters

— n_neighbors → controls local vs global structure

— min_dist → controls clustering tightness

— Metric (distance function)

Types of Unsupervised ML: Density estimation

Goal:

— Estimate the underlying probability density p(x) to understand data structure and identify unusual or rare patterns (anomalies!)

Examples for non-parametric models (no true “learning algorithms”):

— **Histogram methods**:

  — Partition space into discrete bins; count data points per bin.

  — Pros: Simple, fast, interpretable.

  — Cons: Sensitive to bin width & alignment, discontinuous estimate poor in high dimensions

— **Kernel Density Estimation (KDE)**:

  — Smooths each data point using a kernel (e.g., Gaussian) to produce a continuous density estimate

  — Bandwidth controls smoothness (too small → noisy; too large → oversmoothed)

  — Pros: Flexible, smooth, non-parametric

  — Cons: Computationally heavy for large n and high-dimensional data.

Types of Unsupervised ML: Anomaly detection

Goal:

— Identify unusual, rare, or unexpected data points without labels, by learning the structure of “normal” data.

Common Unsupervised Algorithms:

— **Density-Based**: Detect anomalies as low-density observations
  — Histogram DensityKDE, kNN Density
— **Clustering-Based**:
  — k-means: large distance to nearest centroid = anomaly
  — DBSCAN: points not belonging to any cluster = anomaly
— **Boundary & Isolation-Based:**
  — Isolation Forest: anomalies isolate quickly in random trees
  — One-class SVM: defines the “support” region for normal data. Outside = anomaly
— Autoencoder-Based & Deep Probabilistic Models

Applications:

— Fraud detection (banking, e-commerce)
— Network intrusion detection (cybersecurity)
— Manufacturing quality control (detect defective parts)
— Sensor/IoT monitoring (equipment failures)
— Healthcare (rare physiological patterns, abnormal scans)
— Financial risk modelling (unusual market behaviour)

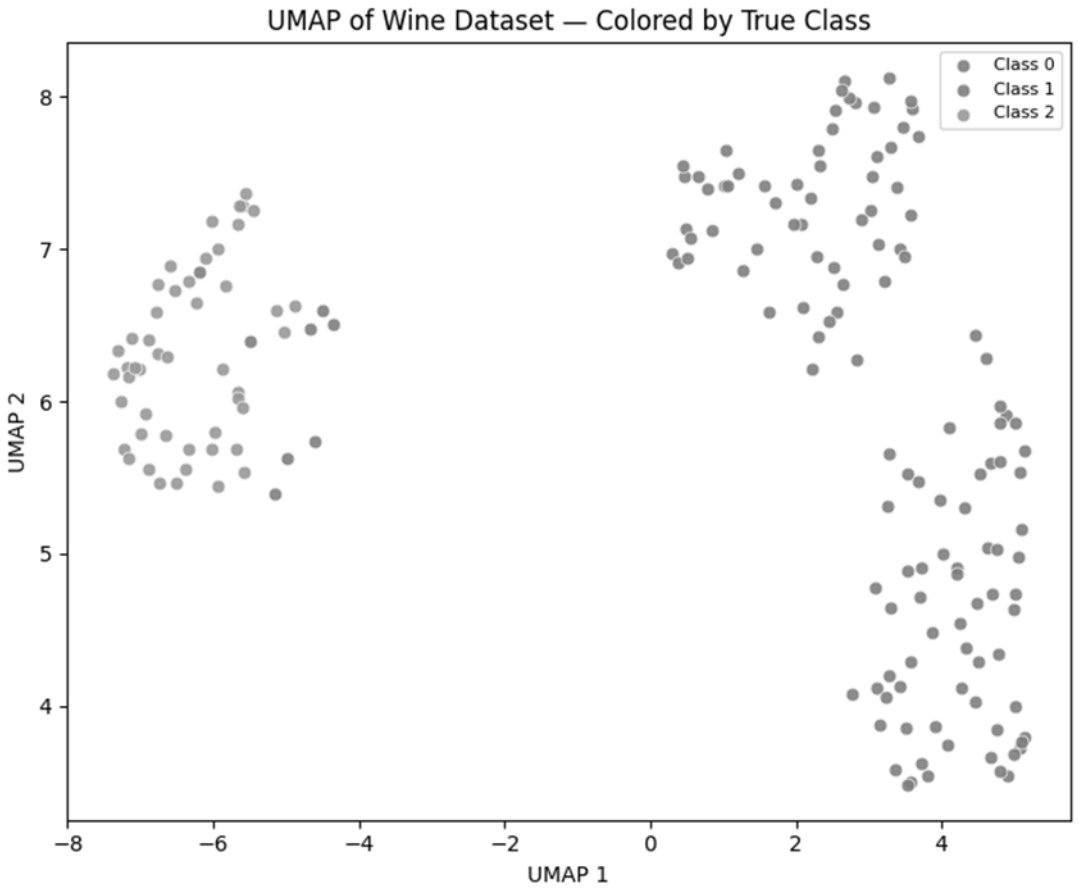
Data Characteristics in Unsupervised ML

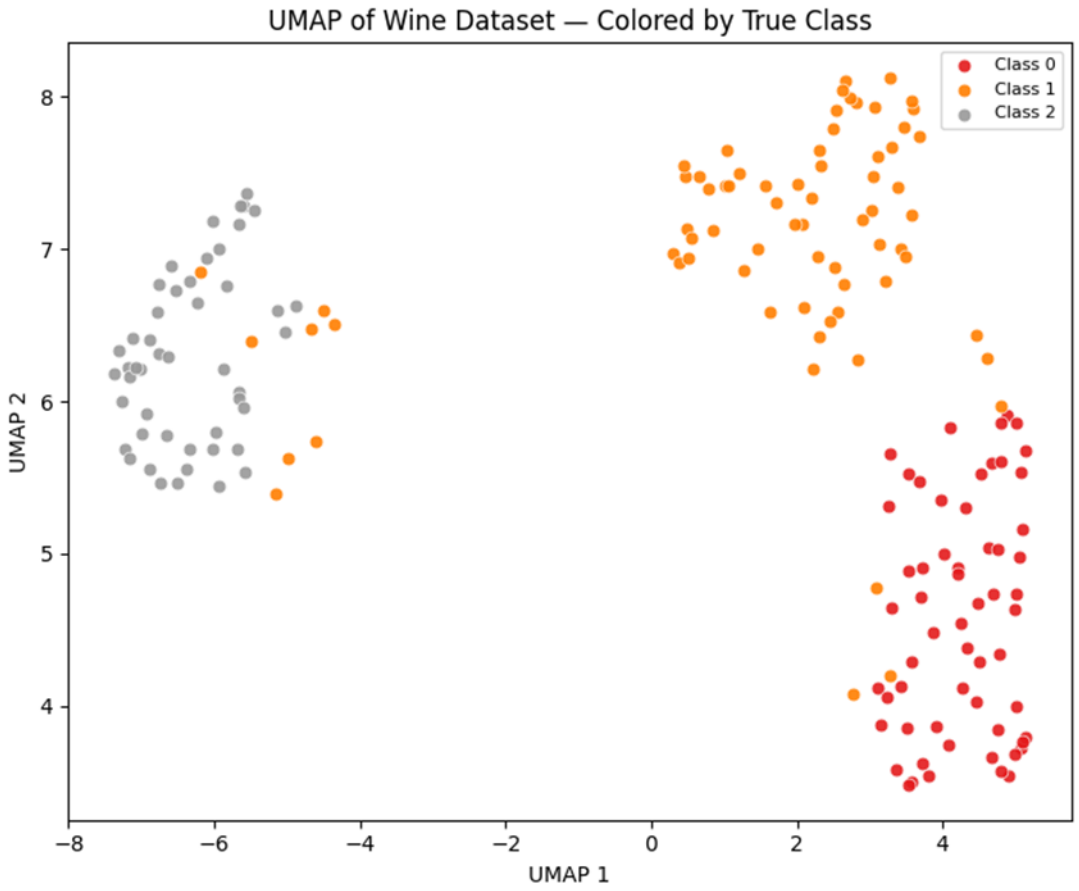
- No label

- Challenges
  - Noise
  - High dimensionality
  - Ambiguous structure

- Preprocessing requirements depends on algorithm (linearity, density, remove irrelevant features (=noise…))
  - Data cleaning (drop or impute missing values, deduplicate)
  - Scaling (almost always needed)
  - **Feature engineering! (Next session!)**
  - Outlier Handling and distribution assessment (e.g. log transformation, sparsity considerations)

Evaluation of Unsupervised ML





Internal evaluation (missing labels!)

External evaluation (requires labelled data)

Relative evaluation (model comparison)

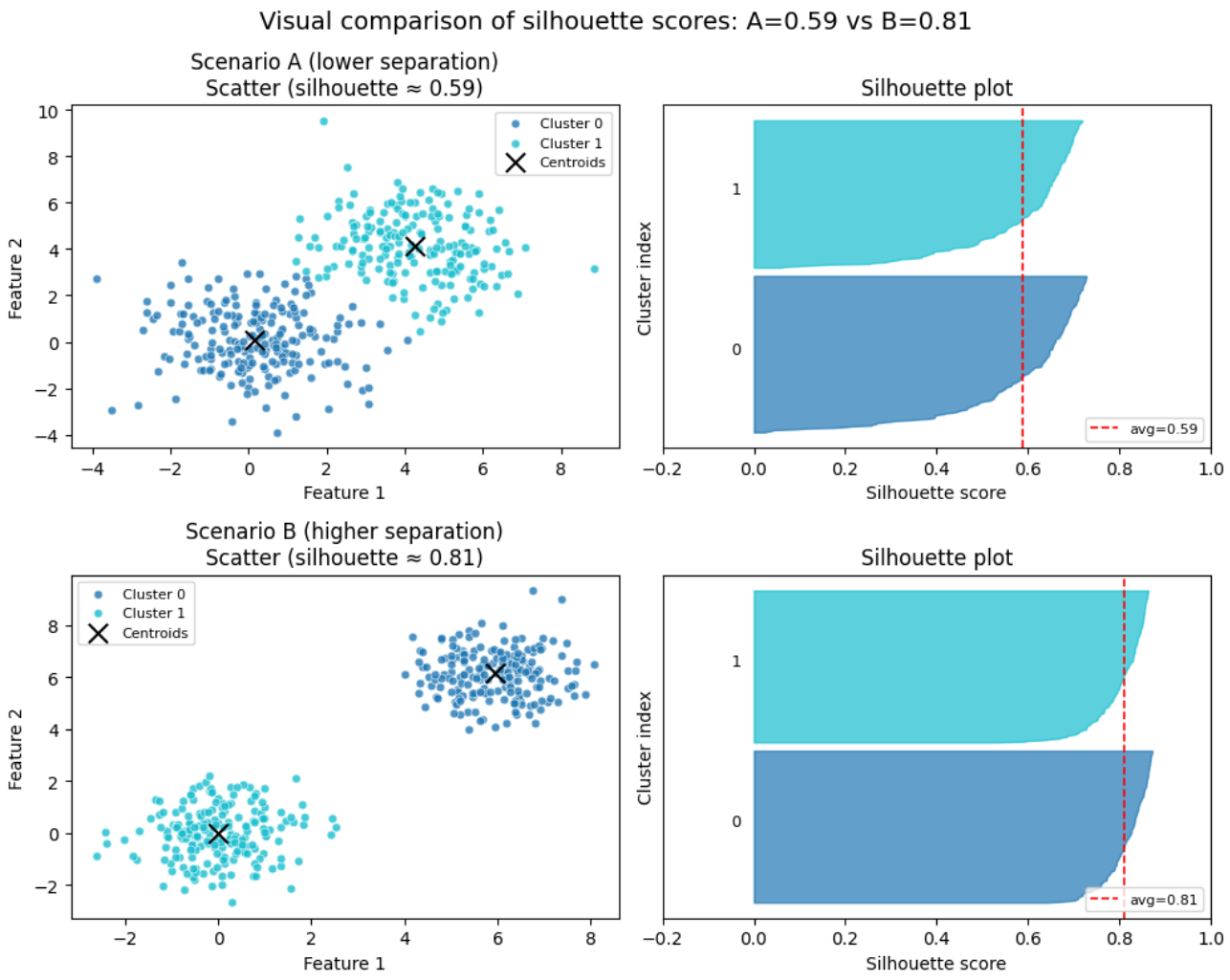
Some recommendations up front:

—Combine multiple metrics
—Validate stability
—Don’t overinterpret clusters
—Prefer domain relevance over metric scores

Evaluation of Unsupervised ML: Internal Metrics

Examples:

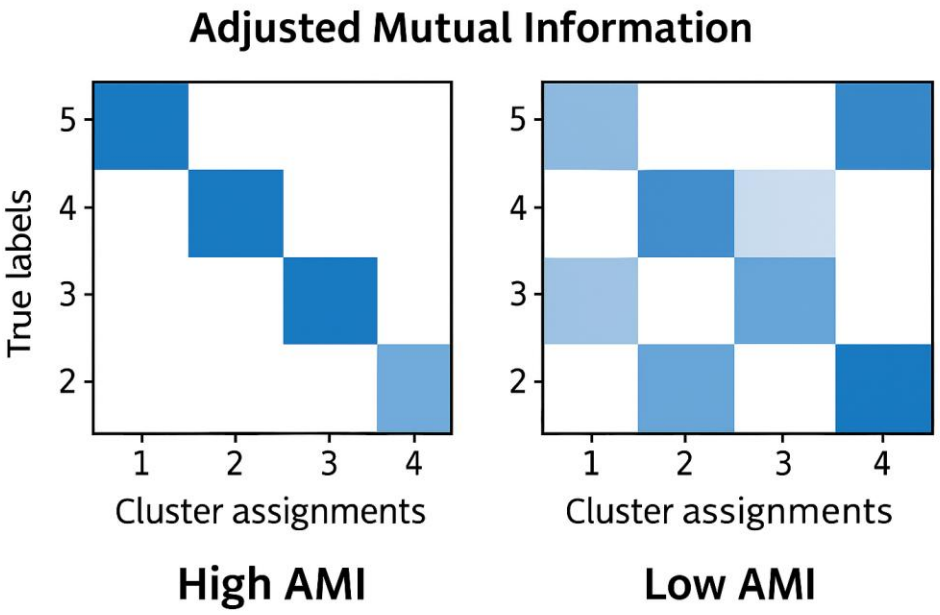
- **Silhouette score:**
  - Measures **how well a data point fits within its assigned cluster**
  - Comparing its average distance to points in the same cluster vs. average distance to points in the nearest other cluster
  - High score: the point is closer to its own cluster than to others = well-separated, compact clusters
- **Davies–Bouldin index:**
  - Measures clustering quality by **comparing how compact** each cluster is **to how far apart** clusters are from each other.
  - Lower values = better clustering, i.e. clusters tight internally and well separated from each other



Evaluation of Unsupervised ML: External Metrics

Examples:

- **Adjusted Rand Index (ARI):**
  - Measures similarity between two clusterings by **counting pairwise agreements** (same cluster vs. different cluster).
  - Ranges from –1 to 1:
    - 1 = perfect match
    - 0 = random labeling
    - < 0 = worse than random
  - Insensitive to label permutations (cluster names don't matter)
- **Adjusted Mutual Information (AMI):**
  - Measures agreement between two clusterings based on **information overlap**
  - Range is 0 to 1 (1 = perfect clustering agreement)
  - Insensitive to label permutations (cluster names don't matter)
  - Works well when clusters differ in size or number
  - **Requires ground-truth labels or a reference clustering**

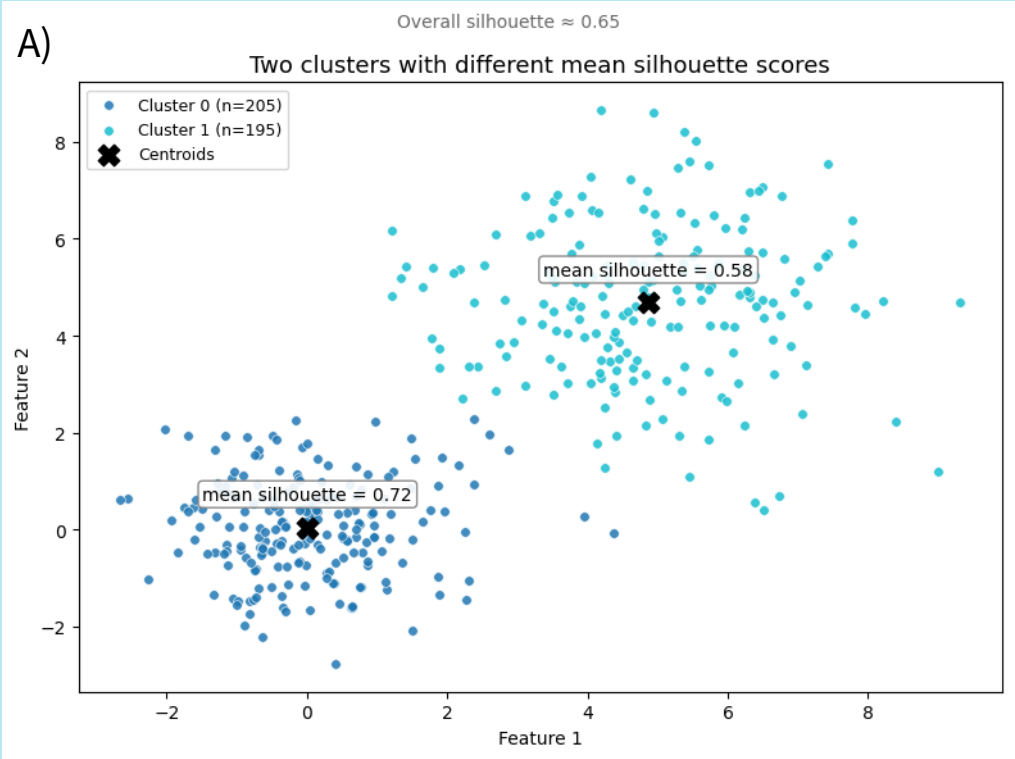


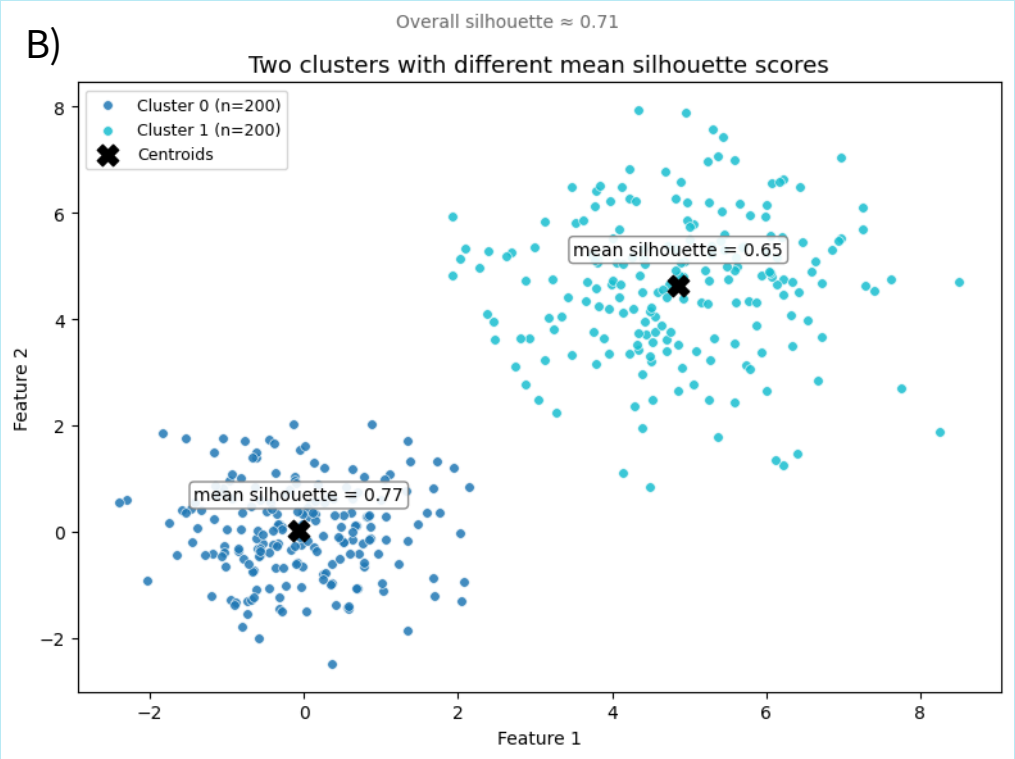
Evaluation of Unsupervised ML: Beyond Metrics

- **Stability analysis**
  - Evaluate consistency of clustering data is perturbed (e.g. subsampling, adding noise)
  - Highlights when cluster boundaries shift heavily
- **Visualisation methods**
  - Compare different clusters side by side, e.g. t-SNE vs UMAP plots of the same dataset
- **Domain expert validation**

Clustering Evaluation Debate

Have a look at the two scenarios (generated example) on the right:





Discuss:

— Which clustering scenario is better?

— Imagine a real-case scenario for these clusters and revisit the first question.

— What if domain knowledge says A is more meaningful?

Clustering: Exploring chemical space

Use the modified ESOL dataset (*esol_modified.csv*) to explore the physicochemical diversity.

1) Data preprocessing
2) Dimensionality reduction: compare PCA, t-SNE, UMAP in visualisation
3) Try KMeans and DBSCAN as clustering methods
4) Evaluate the clustering